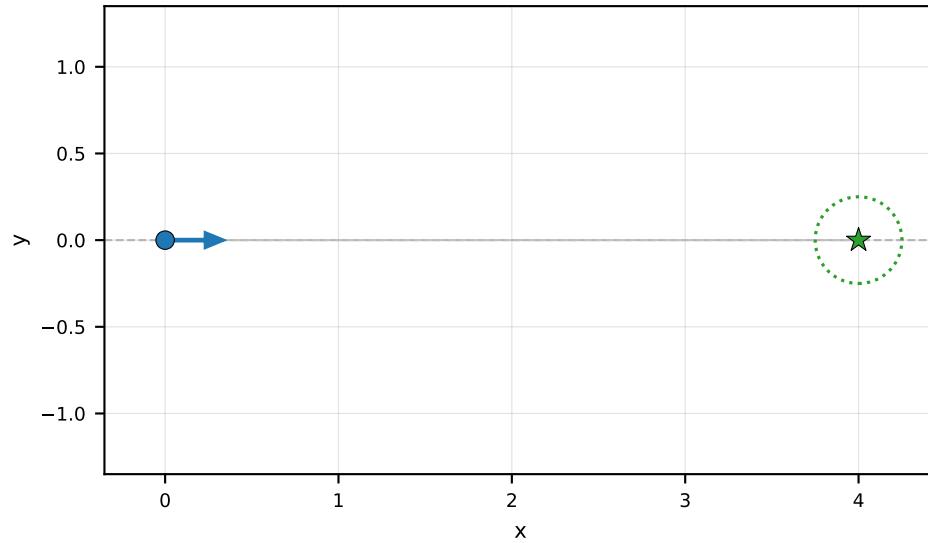


Control and noninterference layouts

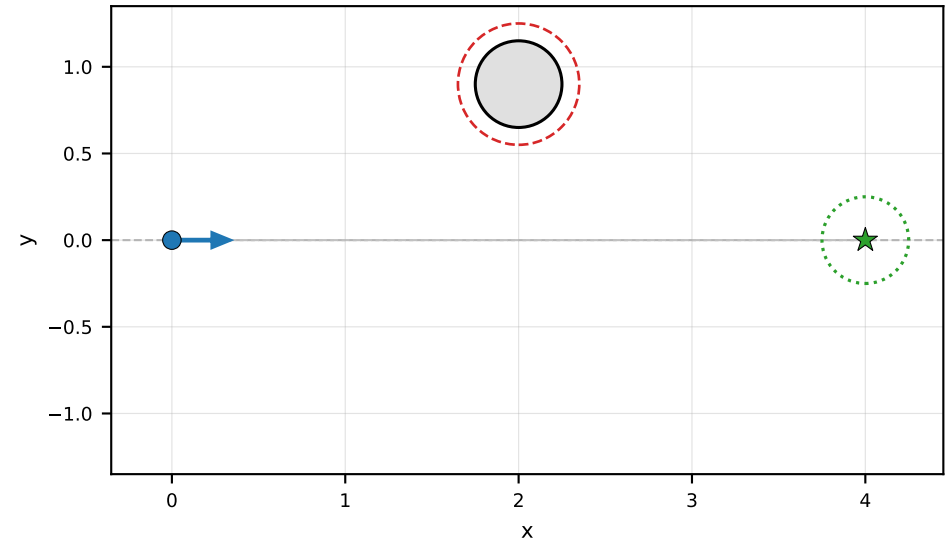
Clear-route cases used to assess basic navigation and noninterference.

Suite: core_navigation_layouts_v1 | Canonical SHA-256: 1027141797052240e83b941398e5a32031e9ca67d001e4a0cf1e19b0f96dd466

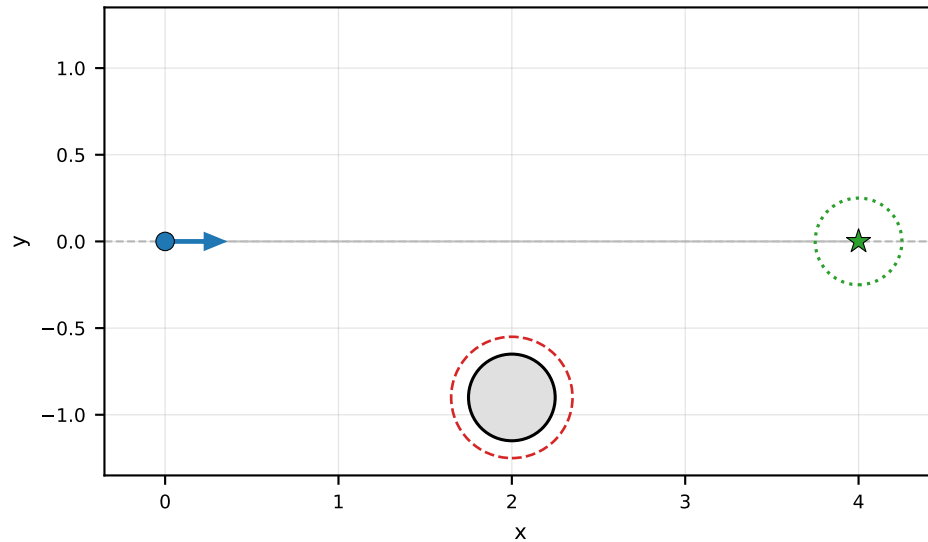
control open route



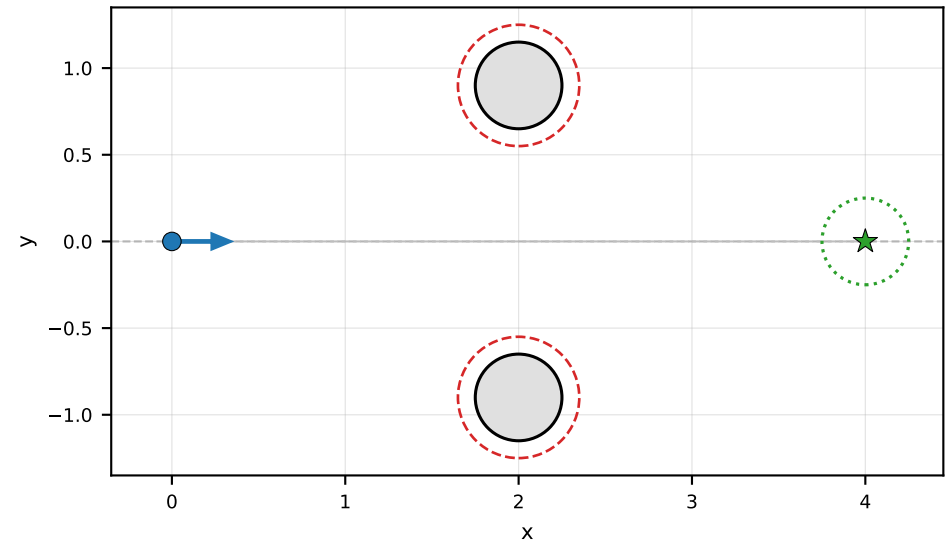
control upper clearance



control lower clearance



control symmetric clearance



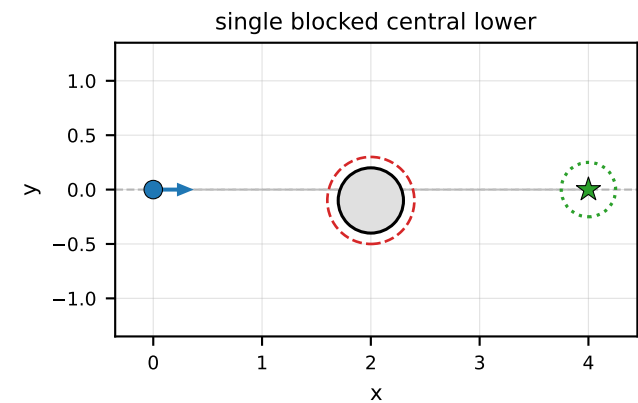
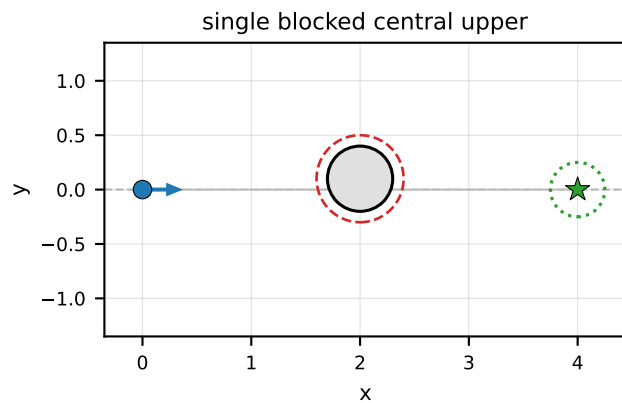
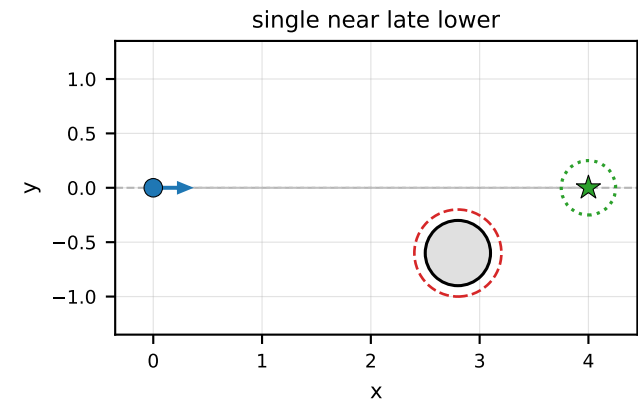
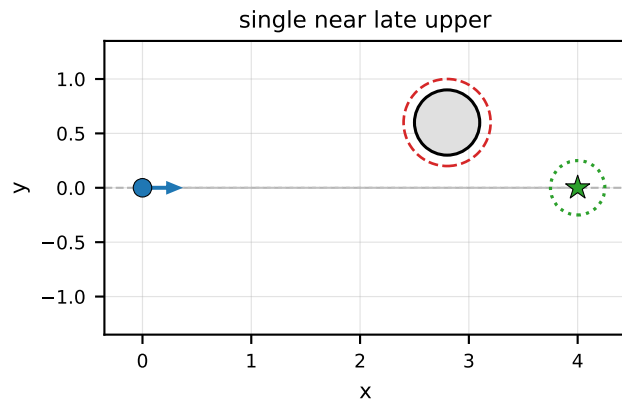
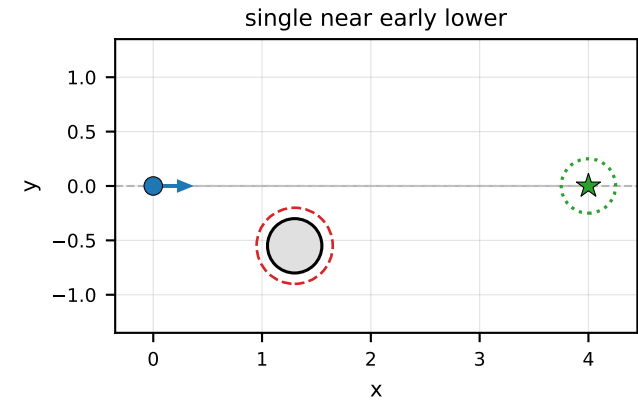
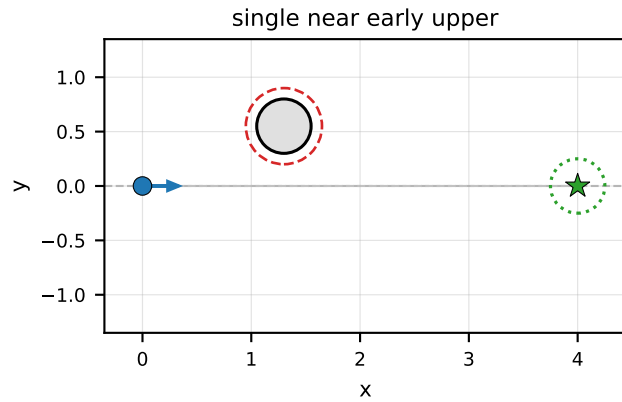
● start and heading ★ goal ○ physical obstacle --- collision boundary --- direct start-goal line

Collision boundary = obstacle radius + agent radius

Single-obstacle deviations

One-obstacle cases varying longitudinal position, lateral offset, and direct-path obstruction.

Suite: core_navigation_layouts_v1 | Canonical SHA-256: 1027141797052240e83b941398e5a32031e9ca67d001e4a0cf1e19b0f96dd466



● start and heading ★ goal ○ physical obstacle - - - collision boundary - - - direct start-goal line

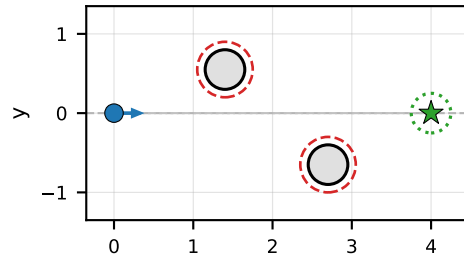
Collision boundary = obstacle radius + agent radius

Two-obstacle interactions

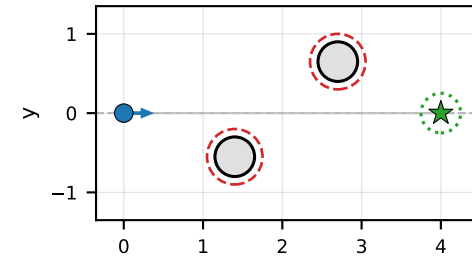
Two-obstacle cases varying staggered and same-side geometry with mirrored counterparts.

Suite: core_navigation_layouts_v1 | Canonical SHA-256: 1027141797052240e83b941398e5a32031e9ca67d001e4a0cf1e19b0f96dd466

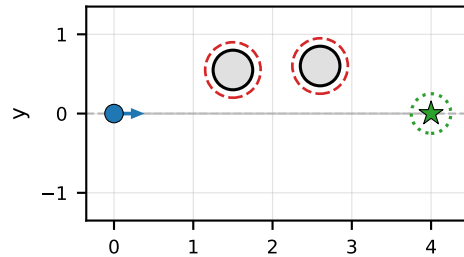
double near staggered upper first



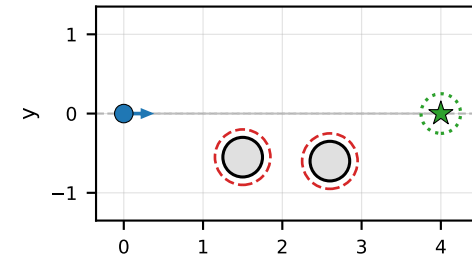
double near staggered lower first



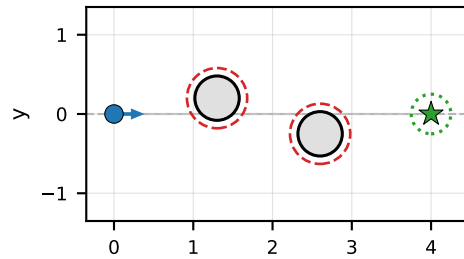
double near same side upper



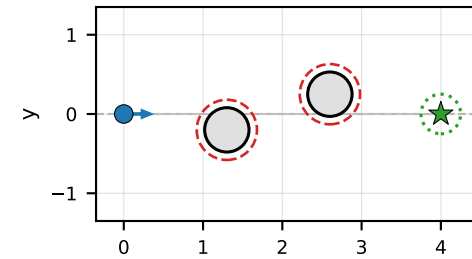
double near same side lower



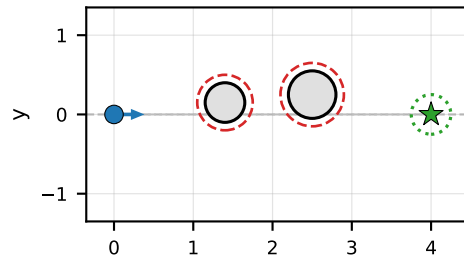
double blocked staggered upper first



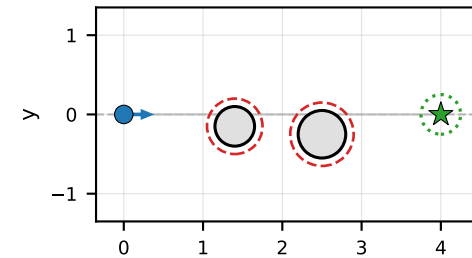
double blocked staggered lower first



double blocked same side upper



double blocked same side lower



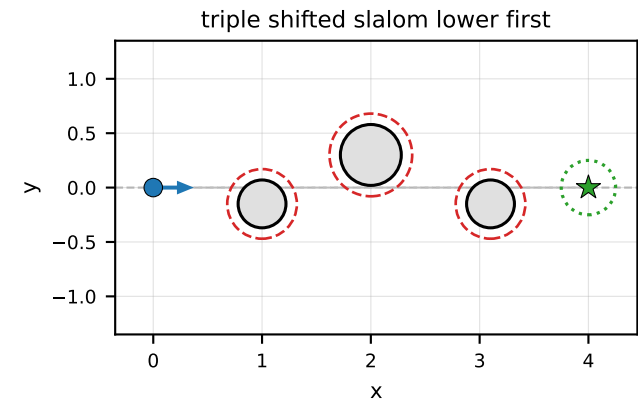
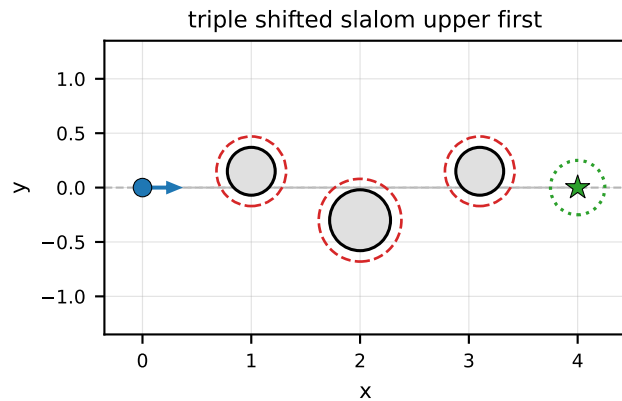
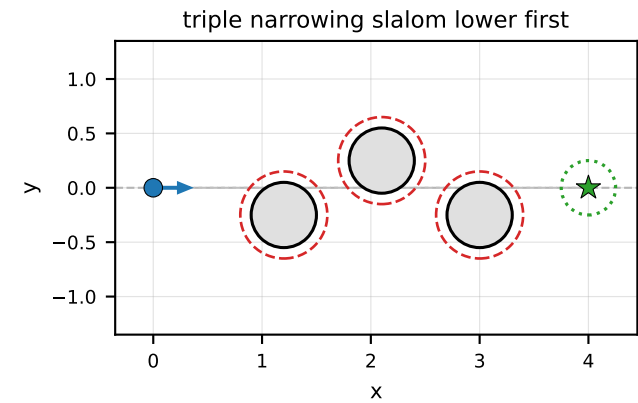
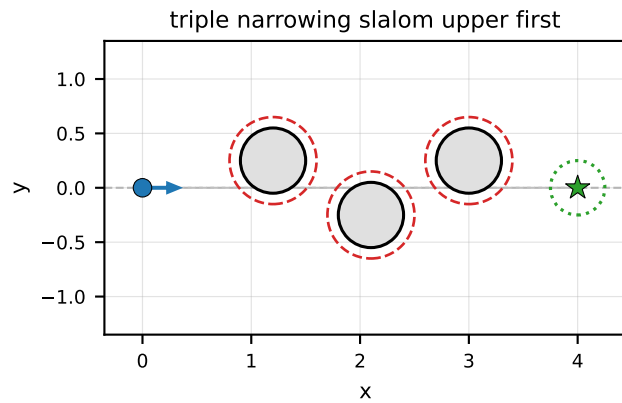
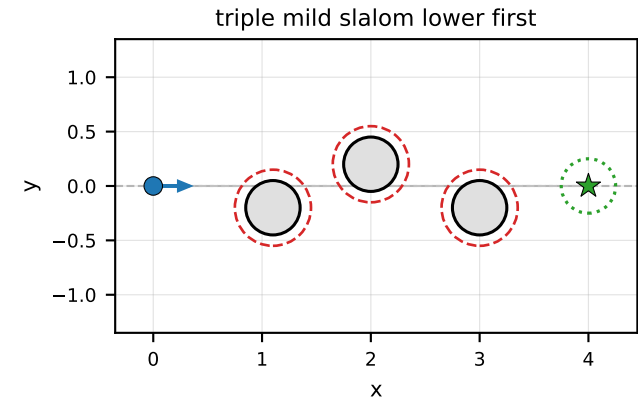
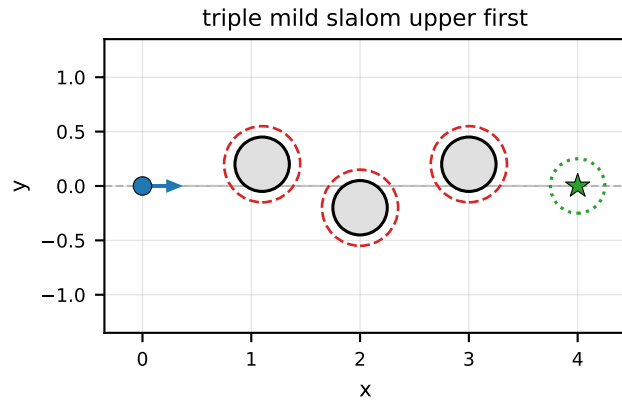
● start and heading ★ goal ○ physical obstacle - - - collision boundary - - - direct start-goal line

Collision boundary = obstacle radius + agent radius

Three-obstacle slaloms

Three-obstacle alternating patterns used to assess sustained avoidance and transfer.

Suite: core_navigation_layouts_v1 | Canonical SHA-256: 1027141797052240e83b941398e5a32031e9ca67d001e4a0cf1e19b0f96dd466



● start and heading ★ goal ○ physical obstacle - - - collision boundary - - - direct start-goal line

Collision boundary = obstacle radius + agent radius